INTRODUCTION TO LaTeX

Phan Thanh Hien et al.

University of Science and Technology of Hanoi

October 2024

Outline

1 Creating a document in LaTeX

Writing your first piece of LaTeX

- The first step is to create a new LaTeX project. You can do this on your own computer by creating a new .tex file; alternatively, you can start a new project in Overleaf.
- Let's start with the simplest working example, which can be opened directly in Overleaf:

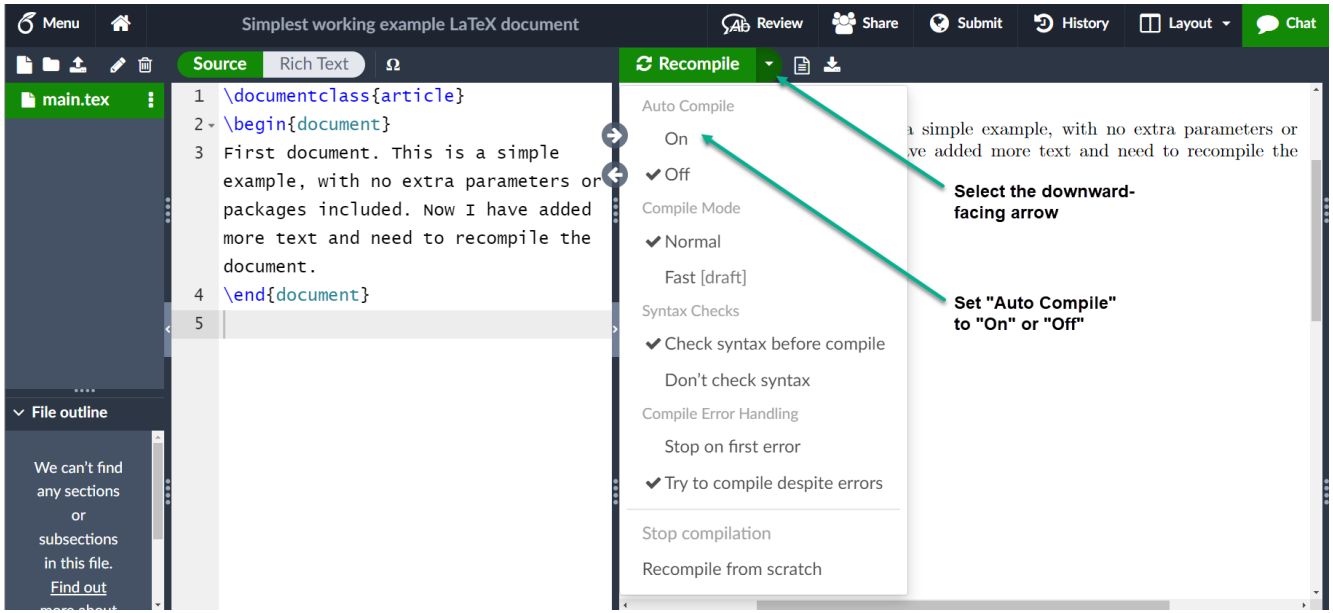
```
\documentclass{article}
\begin{document}
First document. This is a simple
example, with no extra parameters or
packages included. \end{document}
```

First document. This is a simple example, with no extra parameters or packages included.

You can see that LaTeX has automatically indented the first line of the paragraph, taking care of that formatting for you. Let's have a closer look at what each part of our code does.

Recompile

- Any Overleaf project can be configured to recompile automatically each time it is edited: click the small arrow next to the Recompile button and set **Auto Compile** to **On**.



Having seen how to add content to our document, the next step is to give it a title. To do this, we must talk briefly about the preamble.

The preamble of a document

- The previous example showed how document content was entered after the \begin{document} command; however, everything in your **.tex** file appearing *before* that point is called the preamble, which acts as the document's "setup" section.
- Within the preamble you define the document class (type) together with specifics such as languages to be used when writing the document; loading *packages* you would like to use (more on this later), and it is where you'd apply other types of configuration.

```
\documentclass[12pt, a4paper]{article}
\usepackage{graphicx}
```

where `\documentclass[12pt, a4paper]{article}` defines the overall class (type) of document.
The preamble line `\usepackage{graphicx}` is an example of loading an external package (here, **graphicx**) to extend LaTeX's capabilities, enabling it to import external graphics files.

documentclass

Let's have a closer look at what each part of our code does: \documentclass

- declares the document type known as its class, which controls the overall appearance of the document. Different types of documents require different classes

example

- book
- report
- article
- letter
- beamer
- Etc.

You'll be using the 'article' class for your paper, 'beamer' class for your presentation

Including title, author and date information

Adding a title, author and date to our document requires three more lines in the preamble (not the main body of the document). Those lines are:

- \title{My first LaTeX document}: the document title
- \author{Hien PHAN}: here you write the name of the author(s) and, optionally, the \thanks command within the curly braces:
  - \thanks{Funded by the Overleaf team.}: can be added after the name of the author, inside the braces of the author command. It will add a superscript and a footnote with the text inside the braces. Useful if you need to thank an institution in your article.
- \date{October 2024}: you can enter the date manually or use the command \today to typeset the current date every time the document is compiled

```
\documentclass[12pt, a4paper]{article}
\title{My first LaTeX document}
\author{Hien PHAN\thanks{Funded by the Overleaf team.}}
\date{October 2024}
```

Using maketitle command

To typeset the title, author and date use the \maketitle command within the body of the document:

```
\begin{document}
\maketitle{
We have now added a title, author and date to our first \LaTeX{}
document!
\end{document}
```

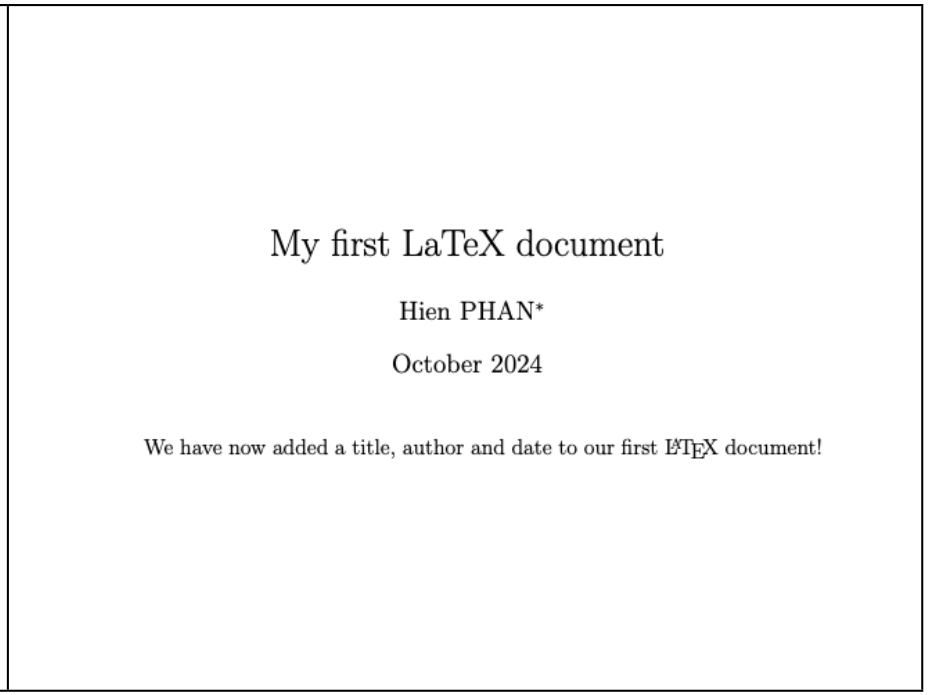
The preamble and body can now be combined to produce a complete document which can be opened in Overleaf:

```
\documentclass[12pt, a4paper]{article}
\title{My first LaTeX document}
\author{Hien PHAN\thanks{Funded by the Overleaf team.}}
\date{October 2024}
\begin{document}
\maketitle{
We have now added a title, author and date to our first \LaTeX{}
document!
\end{document}
```

Maketitle example

This example produces the following output:

```
\documentclass[12pt, a4paper]{article}
\title{My first LaTeX document}
\author{Hien PHAN\thanks{Funded by
the Overleaf team.}}
\date{October 2024}
\begin{document}
\maketitle{
We have now added a title, author and
date to our first \LaTeX{} document!
\end{document}
```



Adding comments

- LaTeX is a form of "program code", but one which specializes in document typesetting; consequently, as with code written in any other programming language, it can be very useful to include comments within your document.
- To make a comment in LaTeX, simply write a **%** symbol at the beginning of the line, as shown in the following code which uses the example above:

```
\documentclass[12pt, a4paper]{article}
\title{My first LaTeX document}
\author{Hien PHAN\thanks{Funded by the Overleaf team.}}
\date{October 2024}
\begin{document}
\maketitle{
We have now added a title, author and date to our first \LaTeX{}
document!
% This line here is a comment. It will not be typeset in the document.
\end{document}
```

Bold, italics and underlining

we will now look at some text formatting commands:

- **Bold**: bold text in LaTeX is typeset using the \textbf{...} command.
- *Italics*: italicised text is produced using the \textit{...} command.
- Underline: to underline text use the \underline{...} command.

Some of the \textbf{greatest} discoveries in \underline{science} were made by \textbf{\textit{accident}}.	Some of the **greatest** discoveries in science were made by ***accident***.

Bold, italics and underlining

Another very useful command is \emph{argument}, whose effect on its *argument* depends on the context. Inside normal text, the emphasized text is italicized, but this behaviour is reversed if used inside an italicized text—see the next example:

```
Some of the greatest \emph{discoveries} in
science were made by accident.

\textit{Some of the greatest
\emph{discoveries} in science were made by
accident.}

\textbf{Some of the greatest
\emph{discoveries} in science were made by
accident.}
```

Some of the greatest *discoveries* in science were made by accident.

Some of the greatest discoveries in science were made by accident.

Some of the greatest *discoveries* in science were made by accident.

Adding images

In this section we will look at how to add images to a LaTeX document. Overleaf supports three ways to insert images:

1. Use the Insert Figure button (), located on the editor toolbar, to insert an image into **Visual Editor** or **Code Editor**.
2. Copy and paste an image into **Visual Editor** or **Code Editor**.
3. Use **Code Editor** to write LaTeX code that inserts a graphic:

```
\documentclass{article}
\usepackage{graphicx} %LaTeX package to
import graphics
\graphicspath{{images/}} %configuring the
graphicx package
\begin{document}
The universe is immense and it seems to be
homogeneous, on a large scale, everywhere we
look.
% The \includegraphics command is provided
(implemented) by the graphicx package
\includegraphics{universe}
There's a picture of a galaxy above.
\end{document
```

The universe is immense and it seems to be homogeneous, on a large scale, everywhere we look.



There's a picture of a galaxy above.

Captions, labels and references

Images can be captioned, labelled and referenced by means of the figure environment, as shown below:

```
\documentclass{article}
\usepackage{graphicx} %LaTeX package to
import graphics
\graphicspath{{images/}} %configuring the
graphicx package
\begin{document}
\begin{figure} [h]
      \centering
      \includegraphics[width=0.5\textwidth]
{mesh}
      \caption{A nice plot.}
      \label{fig:mesh1}
\end{figure}
As you can see in figure \ref{fig:mesh1}, the
function grows near the origin.  This example
is on page \pageref{fig:mesh1}.
\end{document
```

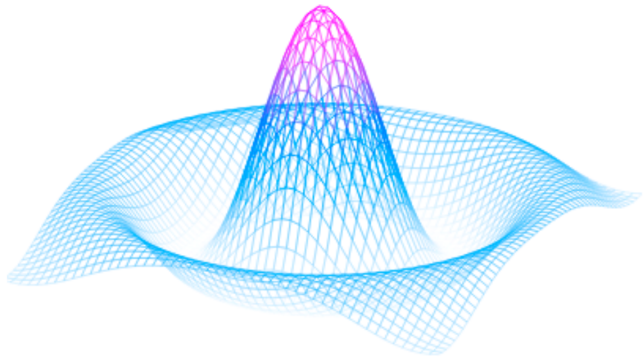


Figure 1: A nice plot.

As you can see in figure 1, the function grows near the origin. This example is on page 1.

Creating lists in LaTeX

- You can create different types of list using *environments*, which are used to encapsulate the LaTeX code required to implement a specific typesetting feature. An environment starts with \begin{environment-name} and ends with \{environment-name} where environment-name might be figure, tabular or one of the list types: itemize for unordered lists or enumerate for ordered lists.
- **Unordered lists** are produced by the itemize environment. Each list entry must be preceded by the \item command, as shown below:

```
\documentclass{article}
\begin{document}
\begin{itemize}
     \item The individual entries are
indicated with a black dot, a so-called
bullet.
     \item The text in the entries may be of
any length.
\end{itemize
\end{document
```

- The individual entries are indicated with a black dot, a so-called bullet.
- The text in the entries may be of any length.

Ordered lists

Ordered lists use the same syntax as unordered lists but are created using the enumerate environment:

```
\documentclass{article}
\begin{document}
\begin{enumerate}
     \item The individual entries are
indicated with a black dot, a so-called
bullet.
     \item The text in the entries may be of
any length.
\end{enumerate
\end{document
```

1. The individual entries are indicated with a black dot, a so-called bullet.
2. The text in the entries may be of any length.

Adding math to LaTeX

- One of the main advantages of LaTeX is the ease with which mathematical expressions can be written. LaTeX provides two writing modes for typesetting mathematics:
  - *inline* math mode used for writing formulas that are part of a paragraph
  - *display* math mode used to write expressions that are not part of a text or paragraph and are typeset on separate lines

Inline math mode::

```
\documentclass[12pt, a4paper]{article}
\begin{document}
In physics, the mass-energy equivalence is
stated by the equation $E=mc^2$, discovered in
1905 by Albert Einstein.
\end{document
```

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Adding math to LaTeX

To typeset inline-mode math you can use one of these delimiter pairs: \(... \), $ ... $ or ..., as demonstrated in the following example: **Inline math mode:**:

```
\documentclass[12pt, a4paper]{article}
\begin{document}
\begin{math}
E=mc^2 is typeset in a paragraph using inline
math mode---as is $E = mc^2$, and so too is
$E = mc^2$.
\end{document
```

$E = mc^2$ is typeset in a paragraph using inline math mode—as is $E = mc^2$, and so too is $E = mc^2$.

Display math mode

Equations typeset in display mode can be numbered or unnumbered, as in the following example:

```
\documentclass[12pt, a4paper]{article}
\begin{document}
In physics, the mass-energy equivalence is
stated by the equation \[E=mc^2\], discovered
in 1905 by Albert Einstein.
In natural units ($c = 1$), the formula
expresses the identity
\begin{equation}
     E=m
\end{equation}
\end{document
```

In physics, the mass-energy equivalence is stated by the equation

$$E = mc^2$$

discovered in 1905 by Albert Einstein. In natural units ($c = 1$), the formula expresses the identity

$$E = m \tag{1}$$

More complete examples

The following examples demonstrate a range of mathematical content typeset using LaTeX.

```
\documentclass[12pt, a4paper]{article}
\begin{document}
Subscripts in math mode are written as $a_b$ and superscripts are written as $a^b$.
These can be combined and nested to write expressions such as
\[ T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q} =
T(x^{i_1},\dots,x^{i_p},e_{j_1},\dots,e_{j_q}) \]
We write integrals using $\int$ and fractions using $\frac{a}{b}$.  Limits are
placed on integrals using superscripts and subscripts:
\[ \int_0^1 \frac{dx}{e^x} = \frac{e-1}{e} \[
Lower case Greek letters are written as $\omega$ $\delta$ etc.  while upper case
Greek letters are written as $\Omega$ $\Delta$.
Mathematical operators are prefixed with a backslash as $\sin(\beta)$,
$\cos(\alpha)$, $\log(x)$ etc.
\end{document
```

More complete examples

The following examples demonstrate a range of mathematical content typeset using LaTeX.

Subscripts in math mode are written as a_b and superscripts are written as a^b. These can be combined and nested to write expressions such as

$$T^{i_1 i_2 \ldots i_p}_{j_1 j_2 \ldots j_q} = T(x^{i_1}, \ldots, x^{i_p}, e_{j_1}, \ldots, e_{j_q})$$

We write integrals using $\int$ and fractions using $\frac{a}{b}$. Limits are placed on integrals using superscripts and subscripts:

$$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$$

Lower case Greek letters are written as ω δ etc. while upper case Greek letters are written as Ω Δ.
Mathematical operators are prefixed with a backslash as $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$ etc.

amsmath equation* environment

The next example uses the equation* environment which is provided by the amsmath package, so we need to add the following line to our document preamble:

```
\usepackage{amsmath}% For the equation* environment
```

amsmath equation* environment

```
\documentclass[12pt, a4paper]{article}
\usepackage{amsmath}% For the equation* environment
\begin{document}
\section{First example}
     The well-known Pythagorean theorem \(x^2 + y^2 = z^2\) was proved to be
invalid for other exponents, meaning the next equation has no integer solutions for
\(n>2\):
     \[ x^n + y^n = z^n \]
\section{Second example}
     This is a simple math expression \(\sqrt{x^2+1}\) inside text.
     And this is also the same:
     \begin{math}
          \sqrt{x^2+1}
     \end{math}
     but by using another command.
     This is a simple math expression without numbering
     \[\sqrt{x^2 + 1}n]
     separated from text.
     This is also the same:
     \begin{displaymath}
          \sqrt{x^2+1}
     \end{displaymath}
     \ldots and this:
     \begin{equation*}
          \sqrt{x^2 + 1}
     \end{equation*}
\end{document
```

More complete examples

First example

The well-known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents, meaning the next equation has no integer solutions for $n > 2$:

$$x^n + y^n = z^n$$

Second example

This is a simple math expression $\sqrt{x^2 + 1}$ inside text.
And this is also the same:
$\sqrt{x^2 + 1}$ but by using another command.
This is a simple math expression without numbering

$$\sqrt{x^2 + 1}$$

separated from text.
This is also the same:

$$\sqrt{x^2 + 1}$$

. . . and this:

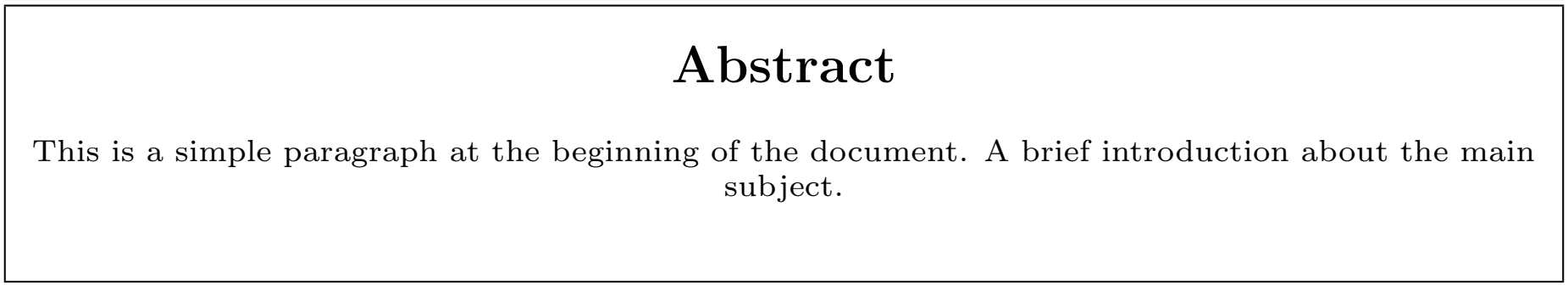
$$\sqrt{x^2 + 1}$$

Basic document structure

Next, we explore abstracts and how to partition a LaTeX document into different chapters, sections and paragraphs.

Abstracts: Scientific articles usually provide an *abstract* which is a brief overview/summary of their core topics, or arguments. The next example demonstrates typesetting an abstract using LaTeX's abstract environment:

```
\documentclass[12pt, a4paper]{article}
\usepackage{amsmath}% For the equation* environment
\begin{document}
\begin{abstract}
This is a simple paragraph at the beginning of the document.  A brief introduction
about the main subject.
\end{abstract
\end{document
```



Paragraphs and new lines

With the abstract in place, we can begin writing our first paragraph. The next example demonstrates:

- how a new paragraph is created by pressing the "enter" key twice, ending the current line and inserting a subsequent blank line;
- how to start a new line without starting a new paragraph by inserting a manual line break using the \\ command, which is a double backslash; alternatively, use the \newline command.

The third paragraph in this example demonstrates use of the commands \\ and \newline:

```
\documentclass[12pt, a4paper]{article}
\begin{document}
\begin{abstract}
This is a simple paragraph at the beginning of the document.  A brief introduction about the
main subject.
\end{abstract
After our abstract we can begin the first paragraph, then press ‘‘enter’’ twice to start the
second one.
This line will start a second paragraph.
I will start the third paragraph and then add \\ a manual line break which causes this text
to start on a new line but remains part of the same paragraph.  Alternatively, I can use the
\verb|\newline|\newline command to start a new line, which is also part of the same
paragraph.
\end{document
```

Paragraphs and new lines

Abstract

This is a simple paragraph at the beginning of the document. A brief introduction about the main subject.

After our abstract we can begin the first paragraph, then press "enter" twice to start the second one.

This line will start a second paragraph.

I will start the third paragraph and then add
a manual line break which causes this text to start on a new line but remains part of the same paragraph. Alternatively, I can use the \newline
command to start a new line, which is also part of the same paragraph.

Note how LaTeX automatically indents paragraphs—except immediately after document headings such as section and subsection—as we will see.

Chapters and sections

Longer documents, irrespective of authoring software, are usually partitioned into parts, chapters, sections, subsections and so forth. Documents created using the book class can be split into parts, chapters, sections, subsections and so forth but the letter class does not provide (support) any commands to do that.

```
\documentclass{book}
\begin{document}
\chapter{First Chapter}
\section{Introduction}
This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortisfacilisis sem.
Nullam nec mi et neque pharetra sollicitudin.  Praesent imperdietmi nec ante.  Donec
ullamcorper, felis non sodales...

\section{Second Section
Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortis facilisissem.
Nullam nec mi et neque pharetra sollicitudin.  Praesent imperdiet mi necante...

\subsection{First Subsection
Praesent imperdietmi nec ante.  Donec ullamcorper, felis non sodales...

\section*{Unnumbered Section
Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortis facilisissem...
\end{document
```

Chapters and sections

Chapter 1

First Chapter

1.1 Introduction

This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortisfacilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

1.2 Second Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

1.2.1 First Subsection

Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

Unnumbered Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem...

Chapters and sections

Collectively, LaTeX document classes provide the following sectioning commands, with specific classes each supporting a relevant subset:

example

- \part{part}
- \chapter{chapter}
- \section{section}
- \subsection{subsection}
- \subsubsection{subsubsection}
- \paragraph{paragraph}
- \subparagraph{subparagraph}

In particular, the \part{part} and \chapter{chapter} commands are only available in the report and book document classes.

Creating tables

Overleaf provides three options to create tables:

1. Using the Insert Table button in the **Visual Editor** (or **Code Editor**) toolbar.
2. *Copying and pasting a table* from another document while using **Visual Editor**.
3. Writing the LaTeX code for the table in **Code Editor**.

If you're new to LaTeX, using the toolbar in **Visual Editor** (option 1) is a great way to get started—you can switch between **Visual Editor** and **Code Editor** to see the code behind the table.

Creating a basic table in LaTeX

We start with an example showing how to typeset a basic table:

```
\begin{center}
\begin{tabular}{c c c}
   cell1 & cell2 & cell3 \\
   cell4 & cell5 & cell6 \\
   cell7 & cell8 & cell9 \\
\end{tabular}
\end{center}
```

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Adding borders

The tabular environment supports horizontal and vertical lines (rules) as part of the table:

- to add horizontal rules, above and below rows, use the \hline command
- to add vertical rules, between columns, use the vertical line parameter |

In this example the argument is {|c|c|c|} which declares three (centred) columns each separated by a vertical line (rule); in addition, we use \hline to place a horizontal rule above the first row and below the final row:

```
\begin{center}
\begin{tabular}{|c|c|c|}
\hline
   cell1 & cell2 & cell3 \\
   cell4 & cell5 & cell6 \\
   cell7 & cell8 & cell9 \\
\hline
\end{tabular}
\end{center}
```

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Adding borders: further example

```
\begin{center}
\begin{tabular}{||c c c c||}
\hline
   Col1 & Col2 & Col2 & Col3 \\ [0.5ex]
   \hline \hline
   1 & 6 & 87837 & 787 \\
   \hline
   2 & 7 & 78 & 5415 \\
   \hline
   3 & 545 & 778 & 7507 \\
   \hline
   4 & 545 & 18744 & 7560 \\
   \hline
   5 & 88 & 788 & 6344 \\ [1ex]
   \hline
\end{tabular}
\end{center}
```

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Captions, labels and references

You can caption and reference tables in much the same way as images. The only difference is that instead of the figure environment, you use the table environment.

```
Table \ref{table:data} shows how to add a
table caption and reference a table.
\begin{table}[h!]
\centering
\begin{tabular}{||c c c c||}
\hline
   Col1 & Col2 & Col2 & Col3 \\ [0.5ex]
   \hline \hline
   1 & 6 & 87837 & 787 \\
   \hline
   2 & 7 & 78 & 5415 \\
   \hline
   3 & 545 & 778 & 7507 \\
   \hline
   4 & 545 & 18744 & 7560 \\
   \hline
   5 & 88 & 788 & 6344 \\ [1ex]
   \hline
\end{tabular}
\caption{Table to test captions and labels.}
\label{table:data}
\end{table}
```

Table 1 shows how to add a table caption and reference a table.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table: Table to test captions and labels.

Adding a Table of Contents

Creating a table of contents is straightforward because the command \tableofcontents does almost all the work for you:

```
\documentclass{book}
\title{Sections and Chapters}
\author{Gubert Farnsworth}
\date{August 2022}
\begin{document}
\maketitle \tableofcontents \section{Introduction}
This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortisfacilisis sem.
Nullam nec mi et neque pharetra sollicitudin.  Praesent imperdietmi nec ante.  Donec
ullamcorper, felis non sodales...

\section*{Unnumbered Section
Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortis facilisissem...

\section{Second Section
Lorem ipsum dolor sit amet, consectetuer adipiscing elit.  Etiam lobortis facilisissem.
Nullam nec mi et neque pharetra sollicitudin.  Praesent imperdiet mi necante...
\end{document
```

Adding a Table of Contents

Sections and Chapters

Gubert Farnsworth

August 2022

Contents

1 Introduction **1**

Unnumbered Section **1**

2 Second Section **1**

1 Introduction

This is the first section.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortisfacilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdietmi nec ante. Donec ullamcorper, felis non sodales...

Unnumbered Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

2 Second Section

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisissem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi necante...

Something more...

- Downloading your finished document
- Finding and using LaTeX packages
- ...

Thank you!

phan-thanh.hien@usth.edu.vn